

A4 Ring Gear Analysis

Note: the excel file is attached with this. Alter the red highlighted cells as needed.

Assumptions

1. A4 is constrained, that is: no axial or radial loading on the motor when it is not rotating
2. Low friction
3. The center of mass of L2, the end effector, and the load are along A4

In short, I assume that A4 has to overcome the inertia of its rotor, L2, EE and the load, but it does not face any significant resisting torques from gravity or friction.

Derivation

Let N = total gear ratio on A4 motor

Let J = total moment of inertia felt by the rotor

From the rotor:

$$T_{\text{rotor}} = J\alpha_{\text{rotor}}$$

The moment of inertia of the load felt by the rotor is reduced by a factor N^2 , thus:

$$T_{\text{rotor}} = \left(J_{\text{rotor}} + \frac{J_{\text{load}}}{N^2} \right) \alpha_{\text{rotor}} \quad (1)$$

Also, due to the gear reduction:

$$\alpha_{\text{L2}} = \frac{\alpha_{\text{rotor}}}{N} \quad (2)$$

$$\alpha_{\text{L2}} = \frac{NT_{\text{rotor}}}{N^2 J_{\text{rotor}} + J_{\text{load}}}$$

Plugging in the Numbers

- We have maximum continuous motor torque $T = 0.0571 \text{ N m}$ (1).
- I cant find the gear ratio of the gearbox, so I estimated $N_{\text{box}} = 50$.
- Using solidworks mass properties, moment of inertia of L2 and EE is ROUGHLY 0.015 kg m^2 .

Plugging in the numbers, I got the following:

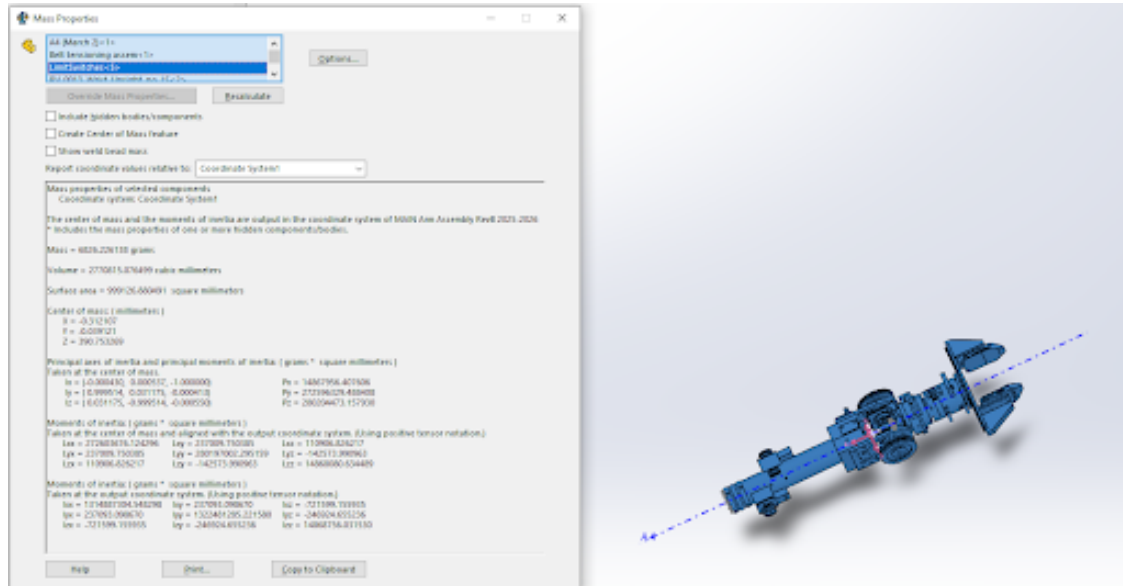


Figure 1: SolidWorks mass properties about the A4 axis.

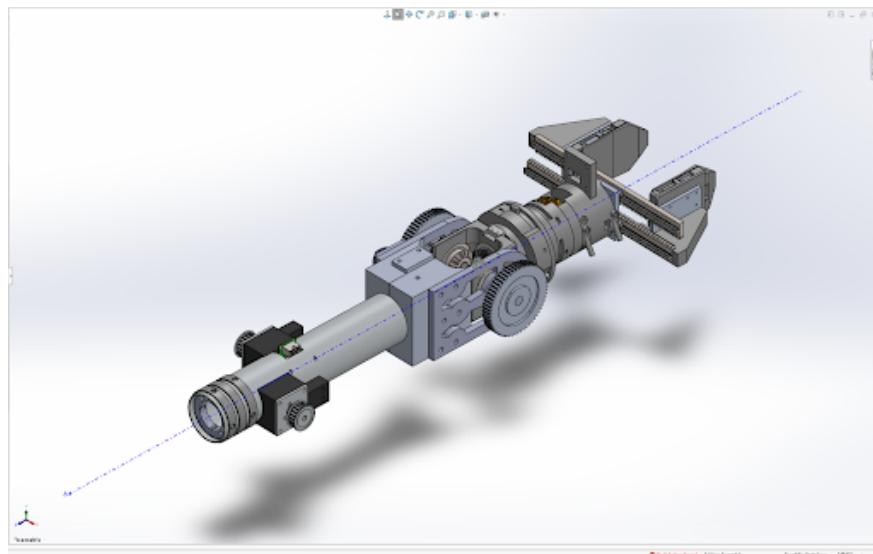


Figure 2: L2 and the end effector aligned with A4.

Motor Torque / Torque on Rotor (T_r)	0.0571 Nm
Load Moment of Inertia (J_L)	0.014868756 kg*m²
Rotor Moment of Inertia (J_r)	0.00000925 kg*m²
Gearbox Gear Ratio	50
Ring Gear Gear Ratio	2
Total Gear Ratio N	100
Maximum Continuous Angular Acceleration WITH RING GEAR:	53.18120663 rad/s²
Maximum Continuous Angular Acceleration WITHOUT RING GEAR:	75.14392622 rad/s²

Figure 3: Excel calculation for the unloaded end effector using $N_{\text{box}} = 50$.

Heavy Pink Steel Rod Test

Next, I gave the EE a heavy pink steel rod (rip A3), and obtained the following:

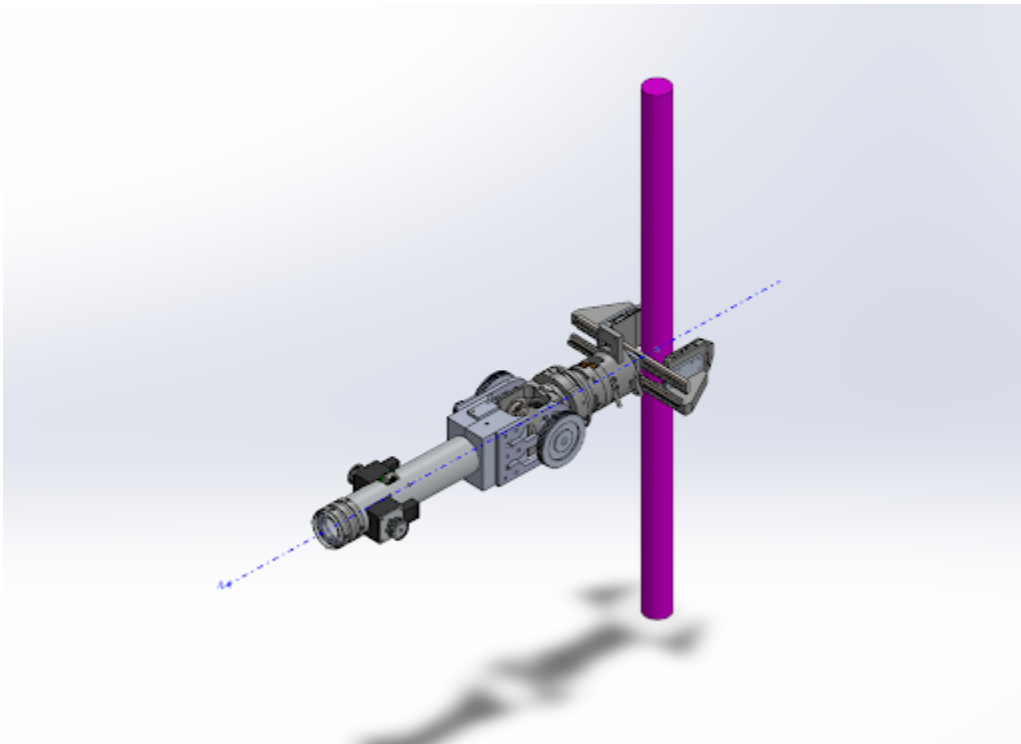


Figure 4: The heavy pink steel rod centered on A4.

Motor Torque / Torque on Rotor (T_r)	0.0571	Nm
Load Moment of Inertia (J_L)	1.293533782	kg*m ²
Rotor Moment of Inertia (J_r)	0.00000925	kg*m ²
Gearbox Gear Ratio	50	
Ring Gear Gear Ratio	2	
Total Gear Ratio N	100	
Maximum Continuous Angular Acceleration WITH RING GEAR:	4.119668709	rad/s ²
Maximum Continuous Angular Acceleration WITHOUT RING GEAR:	2.168367416	rad/s ²

Figure 5: Excel calculation with the heavy steel rod using $N_{\text{box}} = 50$.

Conclusion

As long as $N_{\text{gearbox}} \geq 29$, A4 is actually “quicker” WITHOUT the ring gear when the end effector is unloaded. However, the 2:1 ring gear begins to support A4 angular acceleration at higher loads.

With a known N_{gearbox} , one could easily use the excel calculator to pinpoint the exact load moment of inertia at which the ring gear becomes a useful support (ignoring the extra complexity).

The Dangerous Assumption

Now, the astute among you may realize that assumption 3 is quite dangerous. For example, what if the pole was held like this?:

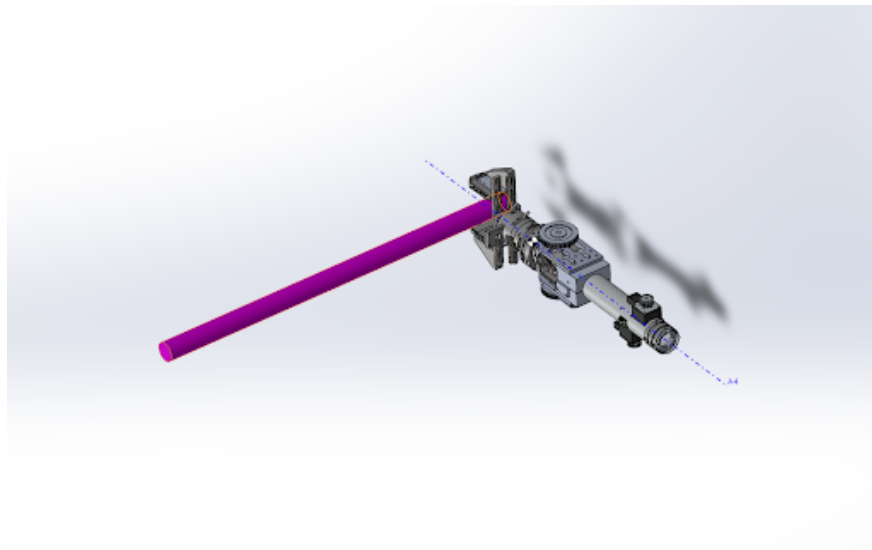


Figure 6: The same rod held asymmetrically about A4.

The rod moves the large center of mass away from A4, creating a resistive gravitational torque on A4. The A4 motor would be cooked.

The math here is pretty simple: by sacrificing the ring gear which has a gear ratio of 2:1, we are exactly reducing the maximum torque output of L2 by this factor of 2. In other words, the maximum asymmetrical load would halve, so to speak.

More specifically:

With a stall torque of 0.263 N m (1), the resistive torque on L2 can be no more than

$$\tau_{L2,\max} = (0.263 \text{ N m}) \times N,$$

where N is the total gear ratio from the motor to L2. For example, after removing the 2:1 ring-gear stage, $N = 50$, so this is about a single apple 9 meters away horizontally from the axis.

TL;DR

By removing the 2:1 ring-gear stage:

Pros:

1. The unloaded A4 angular acceleration is higher, ie A4 rotation is quicker
2. The theoretical maximum A4 angular speed is doubled for the same motor speed
3. Simpler design
4. No more unmeshing of ring gear due to A4 offset
5. Less backlash

Cons:

1. At higher loads, A4 angular acceleration is lower (exact load can be calculated in the attached spreadsheet)
2. The maximum allowable resistive torque on L2 due to friction and gravity is exactly halved. Likely to only be a problem for heavy asymmetric loads, such as distant apples.

Reference:

1. <https://www.maxongroup.com/maxon/view/product/200142>